Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

COMPLICATIONS

YG CG 204 FEB 2022

REVIEW BY: FEB 2025

Diagnosis & Management of Epidural Haematoma Following Neuraxial Anaesthesia

Author(s): Dr Stephan Clements
Reviewer: Departmental Guideline Group

Key Points

- Serious injury to the spinal cord following spinal or epidural blockade is very rare.
- When it occurs, it can result in devastating and often permanent sequelae.
- Literature supports the hypothesis that epidural blockade is of higher risk than spinal blockade.
- Neuraxial procedures performed in the peri-operative period on adults more frequently result in injury compared with procedures in an obstetric, chronic pain or paediatric setting
- If there are any suspicions, early review by a senior anaesthetist, immediate investigations and prompt discussion with the neurosurgical team.

Introduction

This is a neurosurgical emergency. Urgent neurosurgical consultation, for consideration of emergent decompressive laminectomy should be obtained without delay, if suspected.

Diagnosis & Management of Epidural Haematoma

Suspect if there is:

- Sudden onset of severe, constant back pain +/- radiculopathy.
- Back pain that increases with percussion over the spine or coughing and sneezing or straining.
- Return of motor weakness +/- sensory deficit after apparent resolution of the block.

MRI is the diagnostic modality of choice.

Differential Diagnoses:

- Spinal Infection. (Meningitis)
- Spinal epidural abscess. (Very low incidence, insidious)
- Vertebral Canal Haematoma.
- Adhesive Arachnoiditis

Immediate Management: See Flow Chart Below

References

- Anaesthesia/Vol73, Issue S1.January 2018. Spinal Cord Injury arising in anaesthesia practice.
- The 3rd National Audit Project of the Royal College of Anaesthetist.January 2009
- Medical & Clinical Reviews Volume 1 ,Number 1:4 , Neuraxial Anaesthesia Complications.

RED FLAG SYMPTOMS & SIGNS

Prolonged sensory/ motor blockade or outside the expected distribution.

New motor symptoms after regression of block

Unexpected significant back pain: Worse on spine percussion, coughing/straining

Signs and symptoms of Meningitis

Altered bowel or bladder function

ESCALATE

Immediately contact On Call Anaesthetists (Bleep 063 or 551)

May need escalation to Senior Anaesthetist

ASSESSMENT

Urgent review of Medical & Anaesthetic records, Drug & observation chart.

Full Neurological Examination of patient by team

(Including rectal examination for Cauda Equina Syndrome)

If applicable, reassess motor power and sensation every 30 minutes

SCAN & REFER

08:00-20:00 Weekdays: Request Urgent MRI & Contact Neuro-Surgical Registrar, Walton via Switch Board

Out of Hours: Complete On-line Referral Form (https://orioncloud.org/.) & Discuss with Neuro-Surgical Registrar, Walton via Switch Board

If accepted, arrange Blue Light 'Time Critical' WAST Transfer or if Level 2/3 patient, contact Critical Care Desk on 0300 123 2301 to Request ACCTS/EMRTS Transfer